

WHAT IS THIS QUESTION ASKING		
IF THE QUESTION SAYS OR SHOWS	IT IS	GO TO
a shell diagram, or asks for protons, neutrons, valence electrons	atomic structure	2 ATOMS
“electron configuration”, or an ion with a charge	configuration, and how ions differ	2 ATOMS
“molecular shape”, “bond angle”, VSEPR, a formula like BrF ₅	shape from electron pairs	2 ATOMS
“ionic or covalent”, “polar”, electronegativity	bonding and polarity	2 ATOMS
g/mol, “molar mass”, “how many moles”, “how many molecules”	the mole	3 MOLES
a gas with P, V, T, or “at STP”	gas laws	3 MOLES
mol/L, “concentration”, “diluted to”, a burette, a titration	solutions	3 MOLES
“equilibrium”, K _c , K _p , “shifts”, “Le Chatelier”	equilibrium	4 ACIDS
pH, pOH, [H ⁺], acid, base, K _a , buffer	the most common question in the paper	4 ACIDS, the pH tree
“given that...” then two or three equations with ΔH	Hess’s law	5 ENERGY
ΔH, ΔS, ΔG, “spontaneous”, “exothermic”	thermodynamics	5 ENERGY
“rate”, a collision picture, a Maxwell-Boltzmann curve, a catalyst	kinetics	6 ELECTRONS
“oxidation number/state”, “oxidised”, “half-equation”	redox	6 ELECTRONS
two half-cells, a salt bridge, E° values, anode/cathode	electrochemistry	6 ELECTRONS

OUTSIDE THE A105 SYLLABUS		
TOPIC	NOT EXAMINED	WHAT IS USED INSTEAD
Moles and gases	STP and 22.4 L/mol Gay-Lussac the combined gas law	molar volume is 24 dm ³ at 20°C. Two-state problems run PV = nRT twice
Composition	percent composition empirical and molecular formula	absent from A105 entirely
Concentration	ppm % w/v	the four units are % by mass, g/L, mol/kg, mol/L
Equilibrium	ICE tables K _p and partial pressures leaving pure solids out of K _c	every equilibrium one are given is homogeneous
Acids	K _w as a symbol K _a × K _b = K _w the “x is small” approximation pH of a salt	pH + pOH = 14 is given directly. Weak acids are always supplied with a % dissociation
Energy	calorimetry, q = mcΔT	not examined
Cells	cell notation Zn Zn ²⁺ Cu ²⁺ Cu the Nernst equation	everything is at standard conditions
Lesson 11	no deck and no worksheet exists	if it is examinable, the material is not in what one were given – ask

COMMON ERRORS, ALL SIX SIDES		
WHERE	THE TRAP	THE FIX
gases	Celsius into a gas law	K = °C + 273, every time
gases	dm ³ where m ³ is needed	R = 8.314 wants m ³ and Pa
titration	the mole ratio upside down	the bigger coefficient gets the bigger mole count
acids	stopping at pOH	pH = 14 – pOH
acids	the strong-acid method on a weak acid	a ⇌ or a quoted K _a means weak
buffers	the ratio inverted	salt over acid. More salt → pH above pK _a
Hess	reversing without flipping the sign of ΔH	reverse flips the sign, scaling changes the size
Gibbs	kJ mixed with J	ΔS tables are in J. Divide by 1000
Gibbs	–T × (–ΔS) read as minus	two negatives make a plus
redox	the oxidising agent	it is the thing that gets reduced. The two words must be opposites

COMMON ERRORS, ALL SIX SIDES		
WHERE	THE TRAP	THE FIX
cells	multiplying E° when one scale a half-equation	never scale the volts
electrolysis	time in minutes or hours	Q = I × t needs seconds
everywhere	answering to the wrong precision	the energy worksheet asks for 2 decimal places. Show the working either way

READING THE STEM — WHAT EACH INSTRUCTION REQUIRES

HOW THE MARKS DIVIDE

A question ending “**explain your answer**” carries the decision for one mark and a separate reason for each remaining mark. A **4**-mark question therefore needs the answer plus **3** distinct reasons, written as separate sentences.

COMMAND WORDS

“State” and “identify” require the name only. “Explain” requires what happens **and** why it follows. “Show your working” means the arithmetic line itself carries a mark.

Precision: thermodynamics answers are given to **2 decimal places**. Elsewhere match the precision of the data supplied, and state the **unit** every time — several worksheets award a mark for it alone.

UNITS, PRECISION AND THE MARKS ATTACHED TO THEM		
WHERE	CONVENTION	NOTE
gas calculations	V in m ³ , P in Pa, T in K	required by R = 8.314
solution calculations	V in litres for M = n ÷ V	but mL cancels in M.V ₁ = M ₂ V ₂ , and in titration
temperature	K = °C + 273	used throughout, not 273.15
enthalpy	tabulated data kJ/mol, calculated answers kJ	unless an enthalpy of formation is asked for
entropy	J mol ^{–1} K ^{–1}	divide by 1000 before combining with an enthalpy
equilibrium constant	M ^{Δn}	Δn = 0 gives no units
rate constant	M ^(1 – overall order) s ^{–1}	first order is s ^{–1}
electrolysis	t in seconds for Q = It	convert back to hours if asked
thermodynamics answers	2 decimal places	elsewhere, match the precision of the data
every numerical answer	state the unit	several questions award a mark for the unit alone

VSEPR — SHAPE FROM THE ELECTRON PAIRS		
DOMAINS / LONE PAIRS	SHAPE	ANGLE, AND AN EXAMPLE
2 0 lone	Linear	180° CO ₂
3 0 lone	Trigonal planar	120° BF ₃
3 1 lone	Bent or angular	< 120° SO ₂ , O ₃
4 0 lone	Tetrahedral	109.5° CH ₄
4 1 lone	Trigonal pyramid	< 109.5°, about 107.5° NH ₃
4 2 lone	Bent or angular	much < 109.5°, about 104.5° H ₂ O
5 0 lone	Trigonal bipyramid	90° and 120° PCl ₅
5 1 lone	Seesaw (sawhorse)	< 90° and < 120° SF ₄
5 2 lone	T-shape	< 90° ClF ₃
5 3 lone	Linear	180° XeF ₂
6 0 lone	Octahedral	90° SF ₆
6 1 lone	Square pyramid	< 90° BrF ₅
6 2 lone	Square planar	exactly 90° XeF ₄
6 3 lone	T-shape	< 90°

GETTING TO THE SHAPE		
SIGNATURE	METHOD	EXAMPLE
“what is the molecular shape of...”	- draw the Lewis structure - count electron domains on the central atom - count how many of those are lone pairs - read that cell of the table	CO ₂ : two double bonds → 2 domains → linear
what counts as one domain	a double or triple bond is ONE domain , same as a single	counting CO ₂ as 4 gives tetrahedral, which is wrong
a quick route for AX _n with H, F, Cl, Br or I	lone pairs = (valence electrons of A – n) ÷ 2, then domains = n + lone pairs	BrF ₃ : (7–5)/2 = 1 lone, 6 domains → square pyramid
“electron geometry” vs “molecular shape”	geometry is the 0-lone-pair name for that row. Shape is the actual cell	XeF ₄ is octahedral geometry but square planar shape
“will the angles differ?”	lone–lone > lone–bond > bond–bond repulsion. More lone pairs, smaller angle	1 lone → about 107.5° 2 lone → about 104.5° Different, and quote the numbers
a wedge-and-dash drawing	plain line in the plane, bold wedge forwards, dashed backwards	

ATOMS, IONS AND ISOTOPES		
SIGNATURE	METHOD	EXAMPLE
a shell diagram, or a box with blanks	- protons = atomic number, the top number in the box - nuclear charge = +Z, write the plus sign - neutrons = mass number – Z - electrons = Z for a neutral atom only - valence electrons = the outermost ring - configuration fills 2, then 8, then 8	boron: 5p, 6n, mass 11, 2,3, charge 0

ATOMS, IONS AND ISOTOPES		
SIGNATURE	METHOD	EXAMPLE
configuration of an ion	cation loses electrons, anion gains them. Protons and mass do not change	Mg ²⁺ : still 12p and 12n, but 10 electrons → 2,8
identify an atom from a diagram	count the + in the nucleus — that is the element. Then compare electrons to protons for the charge	more electrons than protons → anion
isotope abundances → relative atomic mass	RAM = Σ(isotope mass × its fraction)	Mg: 23.985(0.7899) + 24.986(0.1000) + 25.983(0.1101) = 24.31
counts given, not percentages	divide by the total first	329+829+529 = 1687, so 329/1687 = 19.50%
abundances unknown, RAM given	let x be the lighter fraction, solve one equation	10.013x + 11.009(1–x) = 10.881 → x = 12.88%
balance a nuclear equation	top numbers (mass) balance, bottom numbers (Z) balance, then read the element off Z. alpha is mass 4 / Z 2, beta is mass 0 / Z –1, positron is mass 0 / Z +1	U-238 → ? + alpha mass 238–4 = 234, Z 92–2 = 90 → Th-234
beta or positron?	the notes writes the positron with no plus sign	beta is –1 subscript, positron is +1. Misreading it changes the element

BONDING AND POLARITY		
SIGNATURE	METHOD	EXAMPLE
“what type of bond”	find both electronegativities, subtract, read the band	< 0.5 non-polar covalent 0.5–2.0 polar covalent > 2.0 ionic
worked, all three bands	H–F, K–Cl, Cl–Cl	1.78 polar covalent 2.34 ionic 0 non-polar
“is the MOLECULE polar”	- are the bonds polar? (ΔEN) - what is the shape? - do the bond dipoles cancel?	polar bonds do not make a polar molecule. CO ₂ is linear, so they cancel → non-polar
symmetric AB ₄ with identical B	dipoles cancel → non-polar	change one B, or add a lone pair, and it becomes polar
“which has the highest boiling point”	strongest intermolecular force wins: ion-dipole > hydrogen bond > permanent dipole > temporary dipole	
does it hydrogen bond?	the H must be bonded TO the F, O or N — not merely present in the molecule	CH ₃ F contains F but every H is on carbon → no hydrogen bonding
electronegativity trend	increases across a period, decreases down a group	F is the most electronegative
atomic radius trend	decreases across a period, increases down a group	more shells → valence electrons further out and shielded

REFERENCE DATA NOT PRINTED ON THE PERIODIC TABLE	
Ion charges are not tabulated, and one worksheet question needs it. The periodic table supplied shows atomic number, symbol, name and relative atomic mass — no charges. Group rule: Gp 1 → 1+ Gp 2 → 2+ Gp 13 → 3+ Gp 15 → 3– Gp 16 → 2– Gp 17 → 1– Gp 18 → 0 Then: metal + non-metal is ionic, two non-metals is covalent, and a covalent name tells one the count — trichloride is 3, tetrafluoride is 4. So magnesium chloride is MgCl₂ and aluminium oxide is Al₂O₃. Metallic bonding is not in these two lessons at all — the only three types the notes offers are non-polar covalent, polar covalent and ionic. The notes disagrees with itself twice. The VSEPR image says the tetrahedral angle is 109° while another slide says 109.5° — write 109.5°. And one slide calls relative atomic mass the “nucleon number”, which it is not: relative atomic mass is a weighted average across isotopes, which is why boron is 10.81 and not a whole number.	

WORKED — SHAPE, ANGLE AND POLARITY	
Molecular shape of BrF₅	
Determine the molecular shape and bond angle of BrF ₅ , and state whether the molecule is polar.	
- Valence electrons on the central atom: Br is in Group 17, so 7 - Five are used bonding to F, leaving 7 – 5 = 2 electrons = 1 lone pair - Total electron domains = 5 bonds + 1 lone pair = 6 - Read the 6-domain, 1-lone-pair cell: square pyramid, angle < 90° - Polarity: Br–F bonds are polar (ΔEN = 3.98 – 2.96 = 1.02). The lone pair makes the shape asymmetric, so the dipoles do not cancel	
Square pyramidal, bond angle < 90°, and the molecule is polar	

MODEL WORDING — ATOMS AND BONDING	
BOND ANGLE COMPARISON The bond angle in compound C is smaller , because the central atom has two lone pairs rather than one . Lone pair to lone pair repulsion is greater than lone pair to bond pair repulsion, so the bonding pairs are pushed closer together.	
BOND TYPE The bond is polar covalent , because the difference in electronegativity is 1.78 , which lies between 0.5 and 2.0. The bonding pair is shared unequally between the two atoms.	
COMPARISON, GENERAL SHAPE G:C requires more energy, because it is held by three hydrogen bonds while A:T has only two . More hydrogen bonds make the interaction stronger.	

UNITS AND CONSTANTS		
QUANTITY	UNIT	RELATION
amount	mol	$n = m \div M$
mass, molar mass	g, g/mol	$m = n \times M$
number of particles	atoms or molecules	$N = n \times 6.02 \times 10^{23}$
temperature	kelvin, always	$K = ^\circ C + 273$ (not 273.15)
gas volume, gas pressure	m³ and Pa	required by $R = 8.314 \text{ dm}^3 \cdot \text{Pa} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$ is $\div 1000$, kPa $\rightarrow$ Pa is $\times 1000$
solution volume	L (= dm³)	$M = n \div V$
R, gas constant	8.314 m³·Pa·K⁻¹·mol⁻¹	the only value used; quoted in each question
molar volume	24 dm³/mol at 20°C, 1 atm	recall only. Calculations run through $PV = nRT$
volume chain	1 dm ³ = 1 L = 1000 cm ³ = 1000 mL = 0.001 m³	appears in every part of a gas question

MOLES, MASS AND PARTICLES		
SIGNATURE	METHOD	EXAMPLE
“molar mass of...”	1. count each atom, including subscripts outside brackets 2. sum the relative atomic masses 3. unit is g/mol	CaCO ₃ = 40.08 + 12.01 + 3(16.00) = 100.09 Al ₂ (SO ₄) ₃ : the 3 multiplies S and all four O $\rightarrow$ 342.17
grams $\rightarrow$ moles	$n = m \div M$	102 g HBr $\div$ 80.91 = 1.26 mol
moles $\rightarrow$ grams	$m = n \times M$	0.80 mol KBr $\times$ 119.00 = 95.20 g
“how many molecules”	$N = n \times 6.02 \times 10^{23}$	2.00 mol F ₂ $\rightarrow$ 1.20 $\times 10^{24}$
identify an unknown element	1. get n of the compound from the mole ratio 2. $M = \text{mass} \div n$ 3. subtract the known atoms	$M(\text{XCl}_2) = 110.98$, less 2(35.45) = 40.08 $\rightarrow$ Ca

GASES		
SIGNATURE	METHOD	EXAMPLE
moles change, T and P fixed	Avogadro $V_1/n_1 = V_2/n_2$	5 mol in 50 L $\rightarrow$ 6 mol = 60 L. Only total moles matter
P and V, T fixed	Boyle $P_1V_1 = P_2V_2$	3 atm, 330 mL $\rightarrow$ 1 atm = 990 mL
V and T, P fixed	Charles $V_1/T_1 = V_2/T_2$, T in K	25 $\rightarrow$ 70°C is 298 $\rightarrow$ 343 K. Increase 45/298 = 15.10%
anything else	PV = nRT	$n = \frac{PV}{RT}$ $V = \frac{nRT}{P}$ $T = \frac{PV}{nR}$

GASES		
SIGNATURE	METHOD	EXAMPLE
both P and T change	no combined gas law in A105. Apply $PV = nRT$ twice, or take the ratio	$V_2/V_1 = (P_1/P_2)(T_2/T_1)$
identify a gas from mass and PVT	1. $n = PV/RT$ 2. $M = \text{mass} \div n$ 3. match M to a formula	$M = 0.71/0.0100 = 71 \text{ g/mol} \rightarrow \text{Cl}_2$

WORKED — GAS LAW IN BASE UNITS	
Pressure from PV = nRT	
1.00 mol of nitrogen occupies 11.2 dm ³ at 0 °C. Calculate the pressure in kPa.	
1. Convert temperature: $T = 0 + 273 = \textbf{273 K}$ 2. Convert volume: $11.2 \text{ dm}^3 \div 1000 = \textbf{0.0112 m}^3$ 3. Rearrange: $P = nRT \div V$ 4. Substitute: $P = (1.00)(8.314)(273) \div 0.0112$ 5. $P = 202\,654 \text{ Pa} \div 1000$	
P = 202.65 kPa	

WORKED — MOLAR MASS, MOLES AND PARTICLES	
Three quantities from one sample	
A sample contains 76.00 g of fluorine gas, F ₂ . Calculate its molar mass, the number of moles present, and the number of molecules.	
1. Molar mass: F ₂ has two fluorine atoms, so $M = 2 \times 19.00 = \textbf{38.00 g/mol}$ 2. Moles: $n = m \div M = 76.00 \div 38.00 = \textbf{2.00 mol}$ 3. Molecules: $N = n \times 6.02 \times 10^{23}$ 4. $N = 2.00 \times 6.02 \times 10^{23}$ 5. The question asks for molecules, not atoms. Each F ₂ holds two atoms, so the atom count would be twice this	
$M = 38.00 \text{ g/mol}$ $n = 2.00 \text{ mol}$ $N = \textbf{1.20 \times 10^{24} molecules}$	

WORKED — IDENTIFYING A GAS FROM ITS DENSITY DATA	
Which gas is it	
0.71 g of an unknown gas occupies 250 cm ³ at 101 kPa and 30.7 °C. Identify the gas.	
1. Convert every quantity to base units: $V = 250 \text{ cm}^3 = \textbf{0.00025 m}^3$, $P = 101 \text{ kPa} = \textbf{101\,000 Pa}$, $T = 30.7 + 273 = \textbf{303.7 K}$ 2. Find the moles: $n = PV \div RT$ 3. $n = (101\,000 \times 0.00025) \div (8.314 \times 303.7) = 25.25 \div 2524.9$ 4. $n = \textbf{0.0100 mol}$ 5. Molar mass: $M = \text{mass} \div n = 0.71 \div 0.0100 = \textbf{71 g/mol}$	

6. Match it: Cl ₂ is $2 \times 35.45 = 70.90 \text{ g/mol}$
The gas is chlorine, Cl₂

RELATIVE ATOMIC MASSES IN COMMON USE			
ELEMENT	A _r	ELEMENT	
H hydrogen	1.01	S sulfur	32.07
C carbon	12.01	Cl chlorine	35.45
N nitrogen	14.01	K potassium	39.10
O oxygen	16.00	Ca calcium	40.08
F fluorine	19.00	Fe iron	55.85
Na sodium	22.99	Cu copper	63.55
Mg magnesium	24.31	Zn zinc	65.38
Al aluminium	26.98	Br bromine	79.90
P phosphorus	30.97	I iodine	126.90
He helium	4.00		use 4.00, not 4.003

WORKED — PERCENTAGE CHANGE IN VOLUME	
Volume change when pressure and temperature both alter	
1.5 mol of gas is at 101.3 kPa and 298 K. It is taken to 54.0 kPa and 258 K. Calculate the percentage increase in volume.	
1. There is no combined gas law here , so either apply $PV = nRT$ twice, or take the ratio directly 2. By the ratio: $V_2/V_1 = (P_1 \div P_2) \times (T_2 \div T_1)$ 3. Pressure factor: $101.3 \div 54.0 = 1.876$ 4. Temperature factor: $258 \div 298 = 0.866$ 5. $V_2/V_1 = 1.876 \times 0.866 = \textbf{1.624}$ 6. Percentage increase = $(1.624 - 1) \times 100$	
Volume increases by 62.4%	

WORKED — BALANCING, THEN READING THE MOLE RATIO	
From an unbalanced equation to an amount	
Balance $\text{C}_5\text{H}_{12} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$, then state how many moles of oxygen react with 0.30 mol of C ₅ H ₁₂ .	
1. Balance carbon first: 5 C on the left, so 5CO₂ 2. Balance hydrogen next: 12 H on the left, so 6H₂O 3. Count the oxygen now fixed on the right: 5(2) + 6(1) = 16 atoms 4. Oxygen arrives as O ₂ , so $16 \div 2 = \textbf{8O}_2$ 5. Balanced: $\text{C}_5\text{H}_{12} + \textbf{8O}_2 \rightarrow \textbf{5CO}_2 + \textbf{6H}_2\text{O}$ 6. Ratio $\text{C}_5\text{H}_{12} : \text{O}_2 = 1 : 8$, so 0.30 mol needs 0.30×8	
2.40 mol of oxygen. Balance before reading any ratio	

SOLUTIONS — concentration, dilution, titration, limiting reagent

SOLUTIONS, DILUTION AND TITRATION		
SIGNATURE	METHOD	EXAMPLE
"molarity", mol/L, [X]	$M = n \div V$, V in litres	18.02 g glucose ($M = 180.18$) in 250 mL $\rightarrow$ $0.100/0.250 =$ 0.400 M
the other three units	% by mass = solute $\div$ solution g/L = mass $\div$ volume mol/kg = moles $\div$ solvent in kg	molality divides by the solvent ; the others by the whole solution
"diluted to", a volumetric flask	$M_1V_1 = M_2V_2$	V_1 is the flask capacity , not the water added. 25 mL into a 250 mL flask $\rightarrow V_2 =$ 250
dilution factor, serial dilution	$DF = V_{\text{final}} \div V_{\text{initial}}$	serial dilutions multiply: $10 \times$ $20 = 200$
titration , a burette table	- discard any trial marked Rough - volume used = final – initial, per trial - average the Accurate trials $n = M \times V$ for the known solution - apply the mole ratio, divide by the other volume	the species with the larger coefficient takes the larger mole count
pH given in place of a concentration	$[H^+] = 10^{-\text{pH}}$, for a monoprotic acid this is its molarity	$\text{pH } 1.50 \rightarrow [HCl]$ $= 0.0316 \text{ M}$

WORKED — TITRATION WITH A 2 : 1 RATIO

Concentration of NaOH by titration against H₂SO₄
10.00 mL of NaOH is titrated against 0.0252 mol/L H ₂ SO ₄ . Titres: 15.10, 15.00, 14.90, 15.00 mL, all recorded as accurate. $2 \text{ NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2 \text{ H}_2\text{O}$
- Average the accurate titres: $(15.10 + 15.00 + 14.90 + 15.00) \div 4 =$ 15.00 mL - Moles of the known solution: $n(\text{H}_2\text{SO}_4) = 0.0252 \times 0.01500 \text{ L} =$ $3.78 \times 10^{-4} \text{ mol}$ - Mole ratio from the equation is NaOH : H₂SO₄ = 2 : 1, so multiply by 2 $n(\text{NaOH}) = 2 \times 3.78 \times 10^{-4} = 7.56 \times 10^{-4} \text{ mol}$ - Divide by the NaOH volume: $7.56 \times 10^{-4} \div 0.01000 \text{ L}$
[NaOH] = 0.0756 mol/L

WORKED — LIMITING REAGENT AND PRODUCT MASS

Mass of copper deposited
4.20 g of zinc is added to 200 mL of 0.250 mol/L copper(II) sulfate. $\text{Zn(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu(s)}$. Calculate the mass of copper formed.

- Moles of zinc: $n = 4.20 \div 65.38 =$ 0.0642 mol - Moles of copper(II) sulfate: $n = M \times V = 0.250 \times$ 0.200 L = 0.0500 mol - The ratio is 1 : 1, so each would give that many moles of Cu - CuSO₄ gives the smaller amount, so CuSO₄ is limiting — even though there is less zinc by mass $n(\text{Cu}) = 0.0500 \text{ mol}$, and $M(\text{Cu}) = 63.55 \text{ g/mol}$ $m = 0.0500 \times 63.55$
Mass of copper = 3.18 g

LIMITING REAGENT

SIGNATURE	METHOD	EXAMPLE
two amounts given, or "which is limiting"	- for each reactant, find how much product it alone would give - the smaller product amount identifies the limiting reagent - the other is in excess	4 mol CO (1:1) $\rightarrow$ 4 6 mol H ₂ (2:1) $\rightarrow$ 3 H ₂ gives less $\rightarrow$ H ₂ limiting
"how much remains"	excess = starting amount – amount consumed	0.60 mol H ₂ , 0.40 consumed $\rightarrow$ 0.20 mol left
common error	the smaller starting amount is not necessarily limiting	Zn 0.0642 mol vs CuSO ₄ 0.0500 mol at 1:1 $\rightarrow$ CuSO ₄ limits
one amount given as a solution	convert first: $n = M$ $\times V$	0.250 M $\times$ 0.200 L $= 0.0500 \text{ mol}$

TITRATION TECHNIQUE, AND STANDARDISATION

Why NaOH is standardised: it cannot be weighed accurately because it is hygroscopic and absorbs CO₂ from the air, so its concentration must be found against an acid of known concentration.
Technique: read the meniscus at the bottom ; no air bubble in the burette tip; add dropwise near the endpoint; phenolphthalein as indicator.
MODEL ANSWER — STANDARDISATION The concentration of sodium hydroxide cannot be determined by weighing because it is hygroscopic and absorbs carbon dioxide from the air, so its mass is not reliable. It is therefore standardised against a solution of known concentration .
Outside the syllabus: STP and 22.4 L/mol; Gay-Lussac; the combined gas law; percent composition; empirical and molecular formula; ppm and % w/v.

WORKED — DILUTION FROM A STOCK SOLUTION

Concentration after dilution
25 mL of 6.0 mol/L glucose is pipetted into a 250 mL volumetric flask and made up to the mark. Calculate the moles transferred and the final concentration.

- Moles pipetted: $n = M \times V = 6.0 \times$ 0.025 L = 0.15 mol Dilution adds solvent, not solute, so this 0.15 mol is unchanged by the dilution - Apply $M_1V_1 = M_2V_2$. Here $V_1 = 25 \text{ mL}$ and $V_2 = 250 \text{ mL}$, the flask capacity — not the 225 mL of water added $M_2 = (6.0 \times 25) \div 250$ - Check against the moles: $0.15 \text{ mol} \div 0.250 \text{ L} = 0.60 \text{ mol/L}$, which agrees
$n = 0.15 \text{ mol}$ transferred final concentration = 0.60 mol/L

MODEL WORDING — SOLUTIONS AND TECHNIQUE

STATING A TITRATION RESULT The concentration is 0.0756 mol/L . From the balanced equation the ratio of NaOH to H₂SO₄ is 2 : 1 , so the moles of NaOH are twice those of the acid.
WHY A ROUGH TITRE IS DISCARDED The rough titre is excluded because it is only an approximate value used to locate the endpoint. Averaging the accurate titres gives 15.00 mL .

THE FOUR CONCENTRATION UNITS, SIDE BY SIDE		
UNIT	DIVIDED BY	WORKED
percent by mass, %	mass of the whole solution	0.30 g in 0.30 + 0.50 g $\rightarrow$ 0.30/0.80 = 37.50%
mass concentration, g/L	volume of solution in litres	0.35 g in 1 L $\rightarrow$ 0.35 g/L
molality, mol/kg	mass of the SOLVENT in kg	0.30 mol in 0.50 kg water $\rightarrow$ 0.60 mol/kg
molarity, mol/L	volume of solution in litres	0.30 mol in 0.60 L $\rightarrow$ 0.50 mol/L
the one difference	molality divides by the solvent; every other unit divides by the whole solution	the same 0.30 mol gives four different numbers

WORKED — CONCENTRATION FROM A TITRATION WITH PH DATA
Using pH to obtain the titrant concentration first A burette holds hydrochloric acid of pH 1.50. It is titrated against 25.00 mL of NaOH. Titres: 8.70, 8.80, 8.60, 8.90 mL. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$. Find [NaOH].
- Convert the pH: $[H^+] = 10^{-1.50} =$ 0.0316 mol/L - HCl is monoprotic, so $[HCl] = [H^+] =$ 0.0316 mol/L - Average the titres: $(8.70 + 8.80 + 8.60 + 8.90) \div 4 =$ 8.75 mL - Moles of acid: $n = 0.0316 \times 0.00875 \text{ L} =$ $2.77 \times 10^{-4} \text{ mol}$ - The ratio is 1 : 1, so $n(\text{NaOH})$ is the same - Divide by the NaOH volume: $2.77 \times 10^{-4} \div 0.02500 \text{ L}$
[NaOH] = 0.0111 mol/L

“FIND THE PH OF...” — IDENTIFYING THE CASE		
WHAT IS SHOWN	CASE	ROUTE
single arrow →, a named acid, no K quoted	strong acid	[H ⁺] = (number of H ⁺) × [acid], then pH = −log[H ⁺]
NaOH, KOH, “alkaline”	strong base	two steps. [OH [−]] → pOH = −log[OH [−]] → pH = 14 − pOH
double arrow ⇌, or K _a quoted	weak acid	the % dissociated is required, and is always supplied
a % dissociated figure	weak acid, direct	[H ⁺] = % × [acid], then −log
two solutes, the second a sodium salt of the first	buffer	pH = pK _a + log([salt] ÷ [acid])
weak base with its salt	alkaline buffer	yields pOH. Then pH = 14 − pOH
strong acid with its salt (HCl + NaCl)	not a buffer	treat as the strong acid alone
weak acid with an unrelated salt	not a buffer	treat as the weak acid alone
the two concentrations are equal	pH = pK_a exactly	log(1) = 0; the log term vanishes

PH CONVERSIONS AND STRONG ACIDS		
SIGNATURE	METHOD	EXAMPLE
the four relationships	pH = −log[H ⁺] [H ⁺] = 10 ^{−pH} pOH = −log[OH [−]] [OH [−]] = 10 ^{−pOH} pH + pOH = 14	any quantity reaches any other in at most two steps
monoprotic strong acid: HCl, HNO ₃ , HBr	[H ⁺] = 1 × [acid]	0.03 M HCl → pH = 1.52
diprotic strong acid: H ₂ SO ₄	[H ⁺] = 2 × [acid]	0.015 M → [H ⁺] = 0.030 → pH = 1.52, identical
H₂CO₃, H₂S	diprotic but weak	the doubling applies to strong diprotic acids only
strong base	1. [OH [−]] = [base] 2. pOH = −log[OH [−]] 3. pH = 14 − pOH	0.04 M NaOH → pOH 1.40 → pH 12.60
weak acid, % dissociation supplied	1. [H ⁺] = % × initial concentration 2. pH = −log[H ⁺] 3. K _a = [H ⁺][A [−]] ÷ [HA] if required	0.010 M at 5% → [H ⁺] = 5.0 × 10 ^{−4} → pH 3.30
a weak and a strong acid at the same pH	0.010 M weak and 0.0005 M strong both give 3.30	pH cannot distinguish strong from weak. Only K _a can

BUFFERS		
SIGNATURE	METHOD	EXAMPLE
is the mixture a buffer	1. is the acid (or base) weak ? strong → no 2. are both partners present? acid alone → no 3. is the salt the conjugate of that same acid?	all three conditions must hold

BUFFERS		
SIGNATURE	METHOD	EXAMPLE
acidic buffer	pH = pK_a + log([salt] ÷ [acid])	salt in the numerator. pK _a = −log K _a
alkaline buffer	pOH = pK _b + log([salt] ÷ [base])	yields pOH. Then pH = 14 − pOH
check the direction each time	more salt than acid → ratio > 1 → log positive → pH above pK _a	a result moving the other way indicates an inverted ratio
a small amount of acid is added	the basic partner consumes it	H ⁺ + CH ₃ COO [−] → CH ₃ COOH
a small amount of base is added	the acidic partner consumes it	CH ₃ COOH + OH [−] → H ₂ O + CH ₃ COO [−]
“buffering capacity”	higher concentration of both components, as near equal as possible	two buffers may share a pH and differ greatly in capacity
selecting an acid for a target pH	at equal concentrations pH = pK _a	target 8.6 → the acid with pK _a 8.60
finding a required ratio	[salt] ÷ [acid] = 10 ^(pH − pK_a)	blood: 10 ^(7.40−6.1) → about 20 : 1

WORKED — BUFFER PH, AND THE EFFECT OF ADDING ACID	
pH of a nitrous acid buffer	
A buffer contains 0.80 mol/L HNO ₂ and 0.50 mol/L NaNO ₂ . K _a (HNO ₂) = 4.0 × 10 ^{−4} . Calculate its pH, and state what happens when a little HCl is added.	
1. Convert K _a : pK _a = −log(4.0 × 10 ^{−4}) = 3.40	
2. Identify the parts: acid = HNO ₂ at 0.80 M, salt = NaNO ₂ at 0.50 M	
3. Substitute salt over acid: pH = 3.40 + log(0.50 ÷ 0.80)	
4. log(0.625) = −0.204, so pH = 3.40 − 0.204	
5. Check the direction: less salt than acid, so pH sits below pK _a — consistent	
pH = 3.20 added H ⁺ is consumed by NO ₂ [−] → HNO ₂ , so the pH is nearly unchanged	

EQUILIBRIUM AND K _c		
SIGNATURE	METHOD	EXAMPLE
“write the K _c expression”	1. balance the equation first 2. products in the numerator, reactants in the denominator 3. each concentration raised to its coefficient 4. multiply, never add	CH ₄ + 2H ₂ S ⇌ CS ₂ + 4H ₂ K _c = [CS ₂][H ₂] ⁴ ÷ ([CH ₄][H ₂ S] ²)
“value and units”	units = M ^{Δn} , where Δn = (powers above) − (powers below)	Δn = 0 → no units
Q _c against K _c	identical algebra	Q _c at any instant; equal to K _c only at equilibrium

EQUILIBRIUM AND K _c		
SIGNATURE	METHOD	EXAMPLE
“essentially only product or reactant”	K > 10 ² product K < 10 ^{−2} reactant between: comparable	the boundary is 100. K = 97 is comparable
add reactant, or remove product	shifts right	K _c unchanged
increase pressure	shifts toward fewer moles of gas	equal moles → no shift. K _c unchanged
increase temperature	shifts toward the side that absorbs heat	the only change that alters K_c
catalyst	no shift	no change to position, to K _c , or to yield
heat written into the equation	treat heat as a species	products + heat means exothermic, so heating shifts left

ACIDS — SCOPE NOTES AND MODEL WORDING	
MODEL ANSWER — LE CHATELIER	
The equilibrium shifts to the left , because increasing the temperature favours the endothermic direction, which absorbs the added heat. The amount of CH₄ therefore decreases .	
MODEL ANSWER — BUFFER IDENTIFICATION	
This mixture is a buffer, because it contains a weak acid, HNO₂ , together with its conjugate base, NO₂[−] , supplied by the salt. Added acid is removed by the conjugate base and added alkali by the weak acid, so the pH changes very little.	
Outside the syllabus: ICE tables, K _p and partial pressures, K _w as a symbol, K _a × K _b = K _w , the “x is small” approximation, pH of a weak base from K _b alone, pH of a salt solution, and omitting pure solids from K _c .	
Two printing errors in the notes. One equilibrium equation reads PCl₅ ⇌ PCl₃ + Cl₂ , which is impossible; the left side should be PCl ₃ . Another gives SO₂ + O₂ ⇌ 2SO₃ unbalanced; it should read 2SO ₂ .	
pK_a of ethanoic acid appears once as 4.76 and elsewhere as 4.74 . Use 4.74 .	

WORKED — A WEAK ACID AND A STRONG ACID AT THE SAME PH	
pH, K_a, and what pH cannot tell you	
A 0.010 mol/L solution of the weak acid HY is 5% dissociated. Calculate its pH and K _a . A 0.0005 mol/L solution of the strong acid HX is also measured. Comment on what pH reveals about acid strength.	
1. Dissociated fraction: [H ⁺] = [Y [−]] = 5% × 0.010 = 5.0 × 10^{−4} mol/L	
2. Undissociated acid remaining: [HY] = 95% × 0.010 = 0.0095 mol/L	
3. pH = −log(5.0 × 10 ^{−4}) = 3.30	
4. K _a = [H ⁺][Y [−]] ÷ [HY] = (5.0 × 10 ^{−4}) ² ÷ 0.0095 = 2.6 × 10^{−5}	
5. For the strong acid, dissociation is complete: [H ⁺] = 0.0005, so pH = −log(0.0005) = 3.30	
6. Both solutions have pH 3.30 , yet one acid is weak and the other strong	
pH depends on concentration as well as strength, so it cannot distinguish a weak acid from a strong one. Only K_a can	

HESS'S LAW — COMBINING EQUATIONS		
SIGNATURE	METHOD	EXAMPLE
"calculate $\Delta_{\text{rxn}}\text{H}$ ", then "given that" and two or three equations	- write the target equation - for each species in it, ask whether it is on the correct side - adjust each supplied equation by one of the three moves below - add them and cancel what appears on both sides - if the result is not the target, a move is wrong	the adjusted ΔH values then add up
species already correct	keep the equation	ΔH unchanged
species on the wrong side	reverse the equation	flip the sign of ΔH . Magnitude unchanged
coefficient wrong by a factor	multiply or halve the equation	multiply ΔH by the same factor . Sign unchanged
both faults at once	reverse and scale, independently	reverse gives +966, then halving gives +483

WORKED — HESS'S LAW WITH TWO REVERSALS	
Enthalpy of formation of ammonium chloride	
Calculate $\Delta_{\text{rxn}}\text{H}$ for $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$. Given: (1) $\frac{1}{2}\text{N}_2 + \frac{1}{2}\text{H}_2 \rightarrow \text{NH}_3$, $\Delta\text{H} = -46.1\text{ kJ}$ (2) $\frac{1}{2}\text{H}_2 + \frac{1}{2}\text{Cl}_2 \rightarrow \text{HCl}$, $\Delta\text{H} = -92.3\text{ kJ}$ (3) $\frac{1}{2}\text{N}_2 + 2\text{H}_2 + \frac{1}{2}\text{Cl}_2 \rightarrow \text{NH}_4\text{Cl}$, $\Delta\text{H} = -314.4\text{ kJ}$	
NH_3 must be a reactant, but is a product in (1). Reverse (1) $\rightarrow \Delta\text{H} = +46.1$ HCl must be a reactant, but is a product in (2). Reverse (2) $\rightarrow \Delta\text{H} = +92.3$ NH_4Cl is already a product in (3). Keep (3) $\rightarrow \Delta\text{H} = -314.4$ - No coefficient is wrong, so nothing is scaled - Add: the $\frac{1}{2}\text{N}_2$, 2H_2 and $\frac{1}{2}\text{Cl}_2$ appear on both sides and cancel, leaving $\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4\text{Cl}$ - Sum the enthalpies: $46.1 + 92.3 - 314.4$	
$\Delta_{\text{rxn}}\text{H} = -176.00\text{ kJ}$, exothermic	

ENTHALPY FROM TABLES		
SIGNATURE	METHOD	EXAMPLE
a table of $\Delta_{\text{f}}\text{H}$ values	$\Delta_{\text{rxn}}\text{H} = \Sigma\text{n}(\text{products}) - \Sigma\text{n}(\text{reactants})$	n is the coefficient from the balanced equation
an element appears	its $\Delta_{\text{f}}\text{H}$ is zero	including the diatomics H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , I_2 .
worked	$4\text{FeS}_2 + 11\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3 + 8\text{SO}_2$	$[2(-824) + 8(-297)] - [4(-178) + 0] = -4024 + 712 = -3312.00\text{ kJ}$
a built-in check	the same reaction by the Hess route and the formation route	both give -176.00 kJ ; disagreement indicates an error
a table of bond enthalpies	$\Delta\text{H} = \text{breaking (+)} + \text{forming (-)}$	reactants first , the reverse of the formation route

ENTHALPY FROM TABLES		
SIGNATURE	METHOD	EXAMPLE
worked, combustion of propanone	break $6(\text{C}-\text{H}) + 2(\text{C}-\text{C}) + 1(\text{C}=\text{O}) + 4(\text{O}=\text{O}) = +5957$ form $6(\text{C}=\text{O}) + 6(\text{O}-\text{H}) = -7608$	-1651.00 kJ . A positive combustion result is always wrong
bond counting	each CO_2 holds two $\text{C}=\text{O}$, each H_2O two $\text{O}-\text{H}$, each O_2 one $\text{O}=\text{O}$ to break	3 CO_2 therefore gives 6, not 3

ENTROPY		
SIGNATURE	METHOD	EXAMPLE
"increase or decrease in entropy", no numbers	- count the particles: more on the right raises ΔS - compare states: solid < liquid < aqueous < gas - state the sign	both tests are required – one examined case has no particle change and turns on the state test alone
rapid check	count moles of gas	gas up $\rightarrow$ positive, gas down $\rightarrow$ negative
worked, the case that turns on states	$\text{NaOH}(\text{s}) + \text{acid}(\text{aq}) \rightarrow \text{salt}(\text{aq}) + \text{H}_2\text{O}(\text{l})$. Particles $2 \rightarrow 2$	solid $\rightarrow$ aqueous, so ΔS is positive
a table of S° values	$\Delta\text{S} = \Sigma\text{S}(\text{products}) - \Sigma\text{S}(\text{reactants})$	elements are not zero here . O_2 has $\text{S}^\circ = 205.0$
worked	$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$	$2(193.0) - [191.5 + 3(131.0)] = -198.50\text{ J mol}^{-1}\text{K}^{-1}$

FREE ENERGY AND SPONTANEITY		
SIGNATURE	METHOD	EXAMPLE
ΔH and ΔS both given, and the word spontaneous	- convert T to kelvin: $\text{K} = ^\circ\text{C} + 273$ check the ΔS unit; if J, divide by 1000 $\Delta\text{G} = \Delta\text{H} - \text{T}\Delta\text{S}$ - state the sign and the verdict	$\Delta\text{G} < 0$ spontaneous > 0 not spontaneous $= 0$ equilibrium
the unit trap	one examined question gives ΔH in kJ with ΔS in J ; another gives ΔS already in kJ	read the unit every time
the double negative	$-\text{T} \times (-\Delta\text{S})$ is plus	$-92.2 + 59.30 = -32.90$, not -151.5
worked, ice melting	$\Delta\text{H} = +6.01\text{ kJ}$, $\Delta\text{S} = 22.0\text{ J} = \mathbf{0.0220\text{ kJ}}$, $\text{T} = 298\text{ K}$	$6.01 - (298)(0.0220) = -0.55\text{ kJ/mol}$, spontaneous
"at what temperature"	set $\Delta\text{G} = 0$, so $\text{T} = \Delta\text{H} \div \Delta\text{S}$	$6.01 \div 0.0220 = 273.2\text{ K}$, the melting point of ice
which side of the crossover	both signs $+$ $\rightarrow$ spontaneous above T both $-$ $\rightarrow$ spontaneous below T	the mixed cases never change verdict

WORKED — SPONTANEITY AT TWO TEMPERATURES	
The Haber process	
For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, $\Delta\text{H} = -92.2\text{ kJ/mol}$ and $\Delta\text{S} = -0.199\text{ kJ mol}^{-1}\text{K}^{-1}$. Determine whether the reaction is spontaneous at 298 K and at 773 K.	

ΔS is already in kJ, so no conversion is needed - At 298 K: $\Delta\text{G} = -92.2 - (298)(-0.199)$ Two negatives give a plus: $-92.2 + 59.30 = -32.90\text{ kJ/mol}$, spontaneous - At 773 K: $\Delta\text{G} = -92.2 - (773)(-0.199) = -92.2 + 153.83$ $= +61.63\text{ kJ/mol}$, not spontaneous - Crossover: $\text{T} = \Delta\text{H} \div \Delta\text{S} = -92.2 \div -0.199 = \mathbf{463.3\text{ K}}$, which explains both results
Spontaneous at 298 K, not at 773 K. Both ΔH and ΔS negative $\rightarrow$ spontaneous only below 463.3 K

SPONTANEITY — THE FOUR CASES		
	ΔH NEGATIVE (EXOTHERMIC)	ΔH POSITIVE (ENDOTHERMIC)
ΔS positive	spontaneous at all temperatures	spontaneous at high T, once $\text{T}\Delta\text{S}$ is large enough
ΔS negative	spontaneous at low T , while $\text{T}\Delta\text{S}$ is small	never spontaneous
reason	only the $\text{T}\Delta\text{S}$ term varies with temperature	where ΔH and ΔS agree in sign, temperature cannot change the verdict

ENERGY — UNITS, CONVENTIONS AND MODEL WORDING	
MODEL ANSWER — SPONTANEITY The reaction is spontaneous at 298 K , because $\Delta\text{G} = \Delta\text{H} - \text{T}\Delta\text{S} = -32.90\text{ kJ/mol}$, which is negative . A negative free energy change indicates a spontaneous process.	
MODEL ANSWER — ENTROPY CHANGE Entropy increases , because the number of particles rises from 2 to 3 , and because a gas is produced where there was none. A gas has the greatest number of possible arrangements.	
Units : tabulated data are in kJ/mol ; calculated answers in kJ , unless an enthalpy of formation is asked for, which is per mole. Entropy tables are in J ; divide by 1000 before combining with an enthalpy. Signs : $-\Delta\text{H}$ exothermic, heat released. $+\Delta\text{H}$ endothermic, heat absorbed. Breaking bonds absorbs (+), forming bonds releases (–). "Gives out 81.67 kJ" means $\Delta\text{H} = -81.67$. The minus sign is not written in the question. Thermodynamics answers are given to 2 decimal places . Calorimetry ($q = mc\Delta\text{T}$) is outside the syllabus .	

WORKED — ENTHALPY FROM BOND ENTHALPIES	
Combustion of propanone	
$\text{CH}_3\text{COCH}_3(\text{g}) + 4\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$. Bond enthalpies in kJ/mol: C–H 412, C–C 348, C=O 805, O=O 496, O–H 463.	
- Count the bonds broken. Propanone is $\text{H}_3\text{C}-\text{CO}-\text{CH}_3$: 6 C–H, 2 C–C, 1 C=O. Plus 4 O=O in the oxygen - Breaking = $6(412) + 2(348) + 805 + 4(496) = 2472 + 696 + 805 + 1984 = \mathbf{+5957\text{ kJ}}$ - Count the bonds formed. Each CO_2 holds two C=O, so 3 CO_2 gives 6 C=O. Each H_2O holds two O–H, so 3 H_2O gives 6 O–H - Forming = $6(805) + 6(463) = 4830 + 2778 = -7608\text{ kJ}$ $\Delta\text{H} = \text{breaking} + \text{forming} = 5957 + (-7608)$	
$\Delta_{\text{rxn}}\text{H} = -1651.00\text{ kJ}$. A combustion returning a positive value has the two terms the wrong way round	

KINETICS		
SIGNATURE	METHOD	EXAMPLE
"why does X speed it up"	every reason ends the same way: more successful collisions	surface area, concentration, pressure → more frequent collisions
temperature	the only factor that does both	more frequent collisions and a bigger proportion above E_a
a catalyst	an alternative route with lower E_a	more molecules clear the lower bar. It is not used up, and k increases
a Maxwell-Boltzmann figure	molecules that can react = the area under the curve to the RIGHT of E_a	shade it, then compare the shaded areas
what moves on that figure	temperature shifts the curve right concentration makes it taller, same place catalyst moves the E_a line, not the curve	three different changes. Do not blur them
"rate = $k[A]^m[B]^n$ "	the orders come from experiment , never from the coefficients	one exception: in an elementary step of a mechanism, order = coefficient
find the order from a rates table	find two runs where only one concentration changed. Rate unchanged → zero order doubles → first order quadruples → second	
find the order from half-life	$t_{1/2}$ constant → first order $t_{1/2}$ halves each time → zero order $t_{1/2}$ doubles each time → second	constant 12 s each halving → first order
units of k in 3 seconds	$M^{(1 - \text{overall order})} s^{-1}$	overall 1 → s^{-1} overall 3 → $M^{-2} s^{-1}$
Arrhenius, two conditions	$k = Ae^{-E_a/RT}$, so with A constant $\ln(k_2/k_1) = E_a/RT_1 - E_a/RT_2$	°C → K first. 400°C is 673 K. R is given in kJ, so E_a in kJ goes straight in
higher TEMPERATURE higher CONC. curve SHIFTS right more molecules past E_a curve rises, same spot more molecules past E_a		
Able to react = the area under the curve to the RIGHT of E_a. A catalyst moves the E_a line left; the curve itself does not move.		
The catalyst lowers only the hump. Reactant and product levels do not move, so the enthalpy change is unchanged.		

MODEL WORDING — REDOX, CELLS AND RATE
IDENTIFYING AN AGENT The oxidising agent is MnO₄⁻, because the oxidation state of manganese falls from +7 to +2. It gains electrons and is itself reduced, so it oxidises the other species.
ELECTRON DIRECTION Electrons flow from chromium to nickel, because chromium is the anode, where oxidation releases electrons. They travel through the external circuit to the cathode.
RATE EXPLANATION The rate increases, because raising the temperature increases the proportion of molecules with energy greater than the activation energy, and also increases the frequency of collisions. This increases the number of successful collisions per second.
AGREE OR DISAGREE Yes. The pump moves two different species in opposite directions, which is the definition of an antiporter.

WORKED — ORDER OF REACTION FROM HALF-LIFE
Finding the order, then testing a mechanism The concentration of G falls 0.40, 0.20, 0.10, 0.05, 0.025, 0.0125 mol/L at 0, 12, 24, 36, 48, 60 s. Determine the order with respect to G, and decide which mechanism is consistent: (I) slow step $G + H_2 \rightarrow GH + H$, or (II) slow step $2G \rightarrow G_2$. - Read the half-lives off the table. 0.40 → 0.20 takes 12 s 0.20 → 0.10 takes 12 s; 0.10 → 0.05 takes 12 s - The half-life is constant and does not depend on the starting concentration - A constant half-life is the signature of first order In an elementary step only, the order equals the coefficient. Mechanism I gives rate = $k[G][H_2]$, first order in G - Mechanism II gives rate = $k[G]^2$, second order in G, which contradicts the data First order in G, and mechanism I is the consistent one

WORKED — RATE LAW, AND THE EFFECT OF CHANGING A CONCENTRATION
Rate constant and its units For a reaction, rate = $k[P][Q]^2$. When $[P] = 0.100$ mol/L and $[Q] = 0.400$ mol/L the rate is $0.12 \text{ mol L}^{-1} \text{ s}^{-1}$. State the orders, calculate k with its units, and give the effect of doubling each concentration. - Orders read straight off the exponents: first in P, second in Q, third overall - Rearrange: $k = \text{rate} \div ([P][Q]^2)$ $k = 0.12 \div (0.100 \times 0.400^2) = 0.12 \div 0.0160$ - Units: k carries $M^{(1 - \text{overall order})} s^{-1}$, so for third order that is $M^{-2} s^{-1}$ - Doubling $[P]$: first order, so the rate changes by $2^1 = \mathbf{\times 2}$ - Doubling $[Q]$: second order, so $2^2 = \mathbf{\times 4}$ $k = 7.5 \text{ M}^{-2} \text{ s}^{-1}$ doubling P doubles the rate, doubling Q quadruples it

WORKED — ARRHENIUS WITH A CATALYST
Rate constant at a second temperature

At 400 °C with $E_a = 30 \text{ kJ/mol}$, $k = 3.7 \times 10^{-5} \text{ s}^{-1}$. Find k at 450 °C where a catalyst lowers E_a to 20 kJ/mol. $R = 8.314 \times 10^{-3} \text{ kJ mol}^{-1} \text{ K}^{-1}$, and A is unchanged. Convert to kelvin first: 400 °C = 673 K, 450 °C = 723 K - R is already in kJ, so E_a in kJ/mol goes in without conversion - With A constant, $\ln(k_2/k_1) = E_{a1}/RT_1 - E_{a2}/RT_2$ - First term: $30 \div (8.314 \times 10^{-3} \times 673) = 5.362$ - Second term: $20 \div (8.314 \times 10^{-3} \times 723) = 3.327$ $\ln(k_2/k_1) = 5.362 - 3.327 = 2.034$, so $k_2/k_1 = e^{2.034} = 7.65$ $k_2 = 3.7 \times 10^{-5} \times 7.65$ $k_2 = 2.83 \times 10^{-4} \text{ s}^{-1}$

WHAT CHANGES THE RATE, AND WHY		
FACTOR	MECHANISM	WHICH OF THE TWO
surface area	smaller particles expose more surface, so collisions are more frequent	more collisions
concentration	more molecules per unit volume, so collisions are more frequent	more collisions
pressure (gases)	the same effect as raising concentration	more collisions
temperature	a greater proportion exceed E_a , and collisions become more frequent	both
catalyst	an alternative route of lower E_a , so more molecules clear the barrier	lowers the bar
the sentence that scores	increase in X → increase in [frequency of collisions / proportion above E_a] → more successful collisions → rate increases	reusable for any factor

WORKED — ORDER FROM A TABLE OF INITIAL RATES
Reading two orders off experimental data For $A + B \rightarrow \text{products}$, doubling [A] at fixed [B] doubles the rate. Doubling [B] at fixed [A] leaves the rate unchanged. Write the rate law and give the overall order. - Compare the two runs where only [A] changed: concentration $\times 2$, rate $\times 2$ - Rate $\times 2^1$ when concentration $\times 2$ means first order in A - Compare the two runs where only [B] changed: concentration $\times 2$, rate $\times 1$ - Rate unchanged means zero order in B, so $[B]^0 = 1$ and B does not appear The orders are not the coefficients from the balanced equation; they come only from data - Overall order = $1 + 0$ Rate = $k[A]$, first order overall

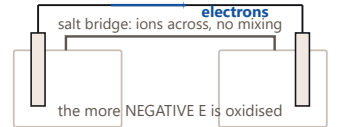
OXIDATION STATES — THE RULES, IN THE ORDER THEY WIN		
ASK, IN THIS ORDER	ASSIGN	EXAMPLE
is it an uncombined element ?	0 , and stop	S ₈ , O ₂ , P ₄ , Na → all 0
is the whole species one atom with a charge ?	= the charge, and stop	Fe ²⁺ → +3
is it Group 1, Group 2, or F ?	+1, +2, −1 — these never bend	Mg(OH) ₂ → Mg = +2 immediately
is it H ?	+1, unless bonded to a metal → −1	NaH, CaH ₂ → H = −1
is it O ?	−2, unless peroxide → −1	H ₂ O ₂ and O ₂ ^{2−} → O = −1
is it Cl, Br or I ?	−1 in a binary compound. If O or F is also there, do NOT assign it — leave it as the unknown	HClO ₄ → Cl is the unknown, not −1
whatever is left	solve: Σ(count × state) = the total charge	neutral → 0 polyatomic ion → its charge
worked	MnO ₄ [−] Mn + 4(−2) = −1	Mn = +7
worked	HClO ₄ (+1) + Cl + 4(−2) = 0	Cl = +7
worked, the neutral-ligand trick	Mn(H ₂ O) ₆ ²⁺ — each H ₂ O is neutral , so contributes 0	Mn = +2
a fast self-check	a state above +7 means one forgot the ion's charge	summing MnO ₄ [−] to 0 gives +8, which is impossible

REDOX — WHO DID WHAT TO WHOM		
SIGNATURE	METHOD	EXAMPLE
“what is oxidised / reduced”	1. assign oxidation states on both sides 2. find the atoms whose number changed 3. went up → oxidised went down → reduced	nothing changed → not a redox reaction
“which is the oxidising agent”	the oxidising agent is the thing that gets REDUCED	the name says what it does to the other species
self-check on the sentence	the two words must be opposites	oxidised → reducing agent reduced → oxidising agent. If they match, it is wrong
balance a redox equation, acid	1. split into two half-equations 2. balance everything except O and H 3. add H₂O for O, then H⁺ for H 4. add electrons to balance the charge 5. scale so the electrons cancel, then add	check: same atoms and same total charge each side

REDOX — WHO DID WHAT TO WHOM		
SIGNATURE	METHOD	EXAMPLE
balance in base	do all of the above via H⁺ first , then add the same number of OH[−] to BOTH sides and combine into H ₂ O	do not start from OH[−] — it is much harder
“will this metal displace that one”	a metal reduces the compounds of anything below it in the activity series	

WORKED — BALANCING A REDOX EQUATION IN ACID	
Dichromate and iodide	
Balance Cr ₂ O ₇ ^{2−} + I [−] → Cr ³⁺ + I ₂ in acidic solution.	
1. Split into half-equations. Iodine: I [−] → I ₂ . Chromium: Cr ₂ O ₇ ^{2−} → Cr ³⁺	
2. Balance atoms other than O and H: 2I[−] → I₂ and Cr₂O₇^{2−} → 2Cr³⁺	
3. Add H ₂ O for oxygen, then H ⁺ for hydrogen: Cr₂O₇^{2−} + 14H⁺ → 2Cr³⁺ + 7H₂O	
4. Add electrons to balance charge. Left is (−2 + 14) = +12, right is +6, so add 6e[−] to the left. Oxidation: 2I[−] → I₂ + 2e[−]	
5. Scale so the electrons cancel: multiply the iodide half by 3 → 6I[−] → 3I₂ + 6e[−]	
6. Add and cancel the electrons. Check charge: left (−2 + 14 − 6) = +6 , right 2(+3) = +6	
Cr₂O₇^{2−} + 14H⁺ + 6I[−] → 2Cr³⁺ + 3I₂ + 7H₂O	

CELLS — GALVANIC AND ELECTROLYTIC		
	GALVANIC (VOLTAIC)	ELECTROLYTIC
how to tell in 3 seconds	no battery — just a voltmeter or bulb	there is a power source in the diagram
the reaction	spontaneous	non-spontaneous, must be driven
E ^o _{cell}	always positive	negative
anode	oxidation, negative	oxidation, positive
cathode	reduction, positive	reduction, negative
what never changes	anode = oxidation, cathode = reduction, in BOTH	electrons flow anode → cathode, in BOTH. Only the sign swaps
the salt bridge	a tube of electrolyte in a gel joining the two half-cells	lets ions flow, stops the solutions mixing
plating or refining	the object one want coated or pure is the cathode	metal always deposits at the cathode



ANODE
oxidation
negative

CATHODE
reduction
positive

Electrons always run anode to cathode.
In an ELECTROLYTIC cell the reactions are the same but the two signs swap over.

WORKED — CELL POTENTIAL AND ELECTRON DIRECTION	
A chromium and nickel voltaic cell	
Two half-cells are joined by a salt bridge. Ni ²⁺ + 2e [−] → Ni, E ^o = −0.28 V. Cr ³⁺ + 3e [−] → Cr, E ^o = −0.74 V. Chromium is on the left. Give the electrode reactions, the overall equation, the cell potential and the direction of electron flow.	
1. Compare the potentials: −0.28 is more positive than −0.74, so Ni²⁺ is reduced and is the cathode	

2. Chromium is therefore oxidised and is the anode . Reverse its equation and flip the sign: Cr → Cr ³⁺ + 3e [−] , E ^o _{ox} = +0.74 V
3. Balance the electrons: the Cr half × 2 and the Ni half × 3 gives 6 electrons each way
4. The E^o values are not scaled. They stay at +0.74 and −0.28
5. E ^o _{cell} = E ^o _{ox} + E ^o _{red} = 0.74 + (−0.28)
6. Electrons leave the anode, which is the chromium on the left , and travel through the external circuit to the cathode on the right
2Cr + 3Ni²⁺ → 2Cr³⁺ + 3Ni E ^o _{cell} = +0.46 V electrons flow left to right

E ^o AND ELECTROLYSIS		
SIGNATURE	METHOD	EXAMPLE
which one is reduced, from two E ^o values	the more positive E ^o is reduced → it is the cathode	“less positive” counts as more negative. +0.34 vs +0.80 → the +0.34 is oxidised
“calculate E ^o _{cell} ”	A105 writes this as E ^o _{cell} = E ^o _{oxidation} + E ^o _{reduction}	reverse the oxidised half and flip the sign of its E ^o . The worksheet’s answer table is printed in this form
is it spontaneous?	E ^o _{cell} positive → yes	negative → no
a half-equation has been scaled	E ^o is NOT multiplied . Volts are per electron	scale the equation, never the volts. ×2 leaves +0.80 as +0.80
worked	Cr (−0.74) with Ni (−0.28)	Cr oxidised: E ^o _{ox} = +0.74 E ^o _{cell} = 0.74 − 0.28 = +0.46 V
electrolysis: mass from current and time	1. Q = I × t, t in seconds 2. moles of electrons = Q ÷ 96500 3. divide by the electrons in the half-equation 4. multiply by the molar mass	2 A for 1 h → Q = 7200 → 0.0746 mol e [−] Na ⁺ + e [−] → Na is 1:1 → 1.72 g
run it backwards for “how long”	moles → ×96500 → Q → ÷ I → seconds → ÷ 3600 for hours	2.5 mol e [−] at 15 A → 4.47 hours
if the question prints an E ^o value	use the question’s value, not the table’s	the tables disagree on Ni (−0.25 vs −0.28)

ELECTRONS — SCOPE NOTES AND ERRATA	
Use the attached potential table , not the short one printed on the slides. They come from different sources and disagree — the attached table is the one the answers follow.	
Not stated in the lecture material: that a catalyst leaves ΔH, the products and the equilibrium position unchanged ; that the salt bridge maintains electrical neutrality and completes the circuit (the notes says only that it lets ions through and stops mixing); what happens when E ^o _{cell} is exactly zero ; and cell notation such as Zn Zn ²⁺ Cu ²⁺ Cu, which A105 does not use.	
Have a periodic table to hand — one electrolysis question needs the molar masses of Hg and Cl and supplies neither.	
Known typos: the activity series prints Mg twice where the second should be Mn ; a car-battery equation prints PbSO ₂ for PbSO₄ ; one hydrogen product is labelled (s) instead of (g); and the iron potential is shown as 0.44 V when it is −0.44 V .	